Mentored Research Grant (MRG)

Project Title

Experiment-Driven Design of Multi-Material 3D-Printed Tensegrity Structures for Energy Absorption

Principal Investigator(s)

PI Name: Jeffrey R. Hill
PI Department: Mechanical Engineering
Co-PI: Sterling G. Baird, Department of Mechanical Engineering

Abstract

We propose a two-year experiment-driven mentored research program in which 2 undergraduate students design, fabricate, and test multi-material 3D-printed tensegrity-inspired structures for energy absorption. Using PLA for rigid struts and TPU for flexible tension elements on a single FDM platform, students will rapidly print and test many candidate geometries, generating high-quality physical data. A Bayesian optimization loop operating directly on experimental measurements will guide the search toward optimal energy-absorbing designs. The project is structured around layered mentoring—weekly one-on-one faculty meetings, lab group sessions, and graduate co-mentoring—that progressively moves students from guided tasks to independent research ownership. Students will gain hands-on experience in CAD design, multi-material 3D printing, mechanical testing, and data-driven optimization. Expected outcomes include undergraduate conference presentations (UCUR, ASME IDETC), a peer-reviewed journal submission, and development of transferable technical skills. This project provides a rich mentored learning environment at the intersection of structural mechanics, advanced manufacturing, and machine learning.

Number of Students to be Mentored

Number of Undergraduate Students: 2
Number of Graduate Students: 1 (co-mentor, in support of creating the mentoring environment)

Budget

Item	Year 1	Year 2	Total
Undergraduate wages	$10,000	$10,000	$20,000
Graduate wages	$1,250	$1,250	$2,500
Supplies	$750	$250	$1,000
Travel	$0	$1,500	$1,500
Total	**$12,000**	**$13,000**	**$25,000**

Relationship to External Funding

The PIs do not currently hold external funding for tensegrity or architected materials research. This MRG provides the primary support for establishing a mentored undergraduate research program in this area. If successful, the preliminary data may inform a future NSF proposal; however, the MRG is self-contained and its outcomes do not depend on external funding.

1 Research Motivation and Overview

We propose a two-year **experiment-driven** mentored research program in which 2 undergraduate students design, fabricate, and test **multi-material 3D-printed tensegrity-inspired structures** for energy absorption. Tensegrity structures—assemblies of rigid compression members suspended within a continuous network of tension members—exhibit remarkable strength-to-weight ratios and tunable mechanical responses, making them attractive for protective equipment, packaging, and aerospace applications [Skelton and de Oliveira, 2009]. Recent work has shown that *tensegrity-inspired* architectures can be directly 3D-printed and retain desirable load-limiting behavior under impact [Pajunen et al., 2019]. Because multimaterial FDM TPU elements are not ideal cables, we target architectures that reproduce key tensegrity-like mechanical responses while remaining manufacturable; notably, TPU's rate-dependent damping is an *asset* for energy absorption rather than merely a deviation from idealized behavior.

Multi-material 3D printing with PLA (rigid struts) and TPU (flexible tension elements) enables rapid fabrication of diverse tensegrity-inspired geometries [Ye et al., 2023, Khatri and Egan, 2024]. Rather than relying on costly simulations, we leverage this manufacturing agility for a **high-throughput experimental approach**: students will print, test, and iterate on many candidate designs—we estimate 50–100+ unique specimens over two years—generating high-quality physical data directly. **Bayesian optimization** (BO) will guide the experimental search, selecting the most informative designs to test next based on prior results [Shahriari et al., 2016, Frazier, 2018]. This experiment-first strategy produces reliable performance data while keeping the project accessible to undergraduates where students can contribute meaningfully through hands-on printing and testing without advanced modeling expertise.

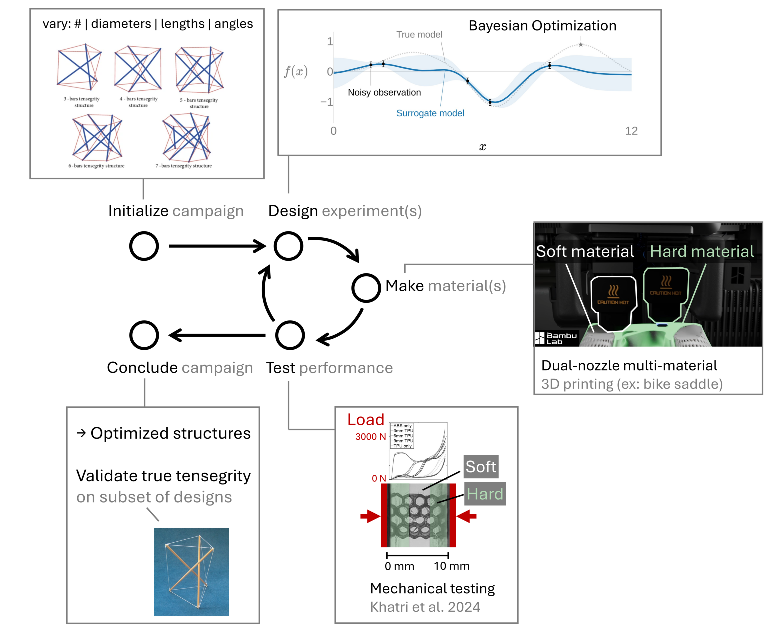

Figure 1: Research framework overview. Students design, print, and test tensegrity-inspired specimens in a closed-loop Bayesian optimization cycle that recommends the next designs to fabricate. Performance is evaluated via quasi-static compression and drop-weight impact tests.

A primary objective is to provide a rich, interdisciplinary mentored research experience for undergraduates. Through weekly one-on-one mentoring, structured lab meetings, and progressive scaffolding from guided tasks to independent research ownership, each student will gain skills in CAD, multi-material 3D printing, mechanical testing, and data-driven optimization. Each student will define design variables, execute experimental campaigns, analyze data, present results, and draft manuscript sections.

2 Background

Tensegrity structures consist of isolated compression members (struts) held in equilibrium by a continuous network of tension members [Skelton and de Oliveira, 2009, Sultan, 2009]. Their defining characteristic—no two rigid bars touch—yields lightweight assemblies with surprising load-bearing capacity and tunable nonlinear responses [Amendola et al., 2014, Zhang et al., 2018]. Pajunen et al. [Pajunen et al., 2019] demonstrated that *tensegrity-inspired* unit cells can be directly 3D-printed and exhibit load-limiting plateaus under drop-weight impact. Multi-material rigid–soft printing [Ye et al., 2023, Khatri and Egan, 2024] has shown that PLA–TPU combinations on a single FDM platform achieve tunable energy absorption via a wrapping-based strategy (rigid cores encapsulated by continuous soft skins) that prevents interface delamination and enables cyclic durability—a critical enabler for our fabrication approach.

Bayesian optimization (BO) is a sequential, model-based strategy for optimizing expensive black-box functions using a Gaussian process surrogate [Shahriari et al., 2016, Frazier, 2018]. BO has driven breakthroughs across domains—from tuning hyperparameters of AlphaGo [Silver et al., 2016] to optimizing sustainable concrete formulations [Ament et al., 2023] and discovering organic laser emitters [Strieth-Kalthoff et al., 2024]—and has been applied to metamaterial design [Mo et al., 2023]. In our framework, BO operates *directly on experimental data*: each round of printing and testing updates a Gaussian process surrogate, which recommends the next designs to fabricate. This closed-loop print–test–optimize cycle eliminates the need for simulation and keeps the workflow centered on physical experimentation—naturally suited to mentoring undergraduates.

3 Student Research Project 1: Design, Fabrication, and Testing

Student profile: One sophomore or junior undergraduate. No prior simulation or advanced modeling experience is required; the hands-on nature of this project makes it an ideal entry point for students early in their studies.

Scope and Deliverables

This student will lead the design-and-build cycle that forms the experimental backbone of the project. Working under weekly mentoring from Prof. Hill, the student will **parameterize a family of tensegrity-inspired unit cells** in CAD (SolidWorks or Fusion 360), defining key variables—strut diameter and length (PLA), tension-element cross-section (TPU), connectivity topology, and unit-cell tiling—that span the design space. Following Ye et al. [2023], the designs use a core-wrapping strategy (rigid PLA struts encapsulated by continuous TPU skins) to ensure robust mechanical interlocking. PLA–TPU interface robustness will be validated during early print trials.

The student will then **fabricate structures** using the department's multi-material FDM printer, iterating on print parameters to achieve reliable geometries. The student works closely with Student 2, who tests each batch and returns performance data. Prof. Baird's master's student serves as a day-to-day co-mentor, guiding equipment operation and troubleshooting.

Mentoring outcomes: The student will develop skills in CAD, multi-material additive manufacturing, and systematic experimental design. Structured mentoring scaffolds move the student from guided replication of existing designs to independent creation of novel geometries by mid-Year 1.

The student will present results at **UCUR** or **ASME IDETC** and contribute as co-author on a journal manuscript.

4 Student Research Project 2: Mechanical Testing and Data-Driven Optimization

Student profile: One sophomore, junior, or senior undergraduate with coursework or interest in mechanics of materials or data analysis.

Scope and Deliverables

This student will lead physical testing and the Bayesian optimization campaign, working under weekly one-on-one mentoring from Prof. Baird. The work begins with **experimental testing** on structures fabricated by Student 1: quasi-static compaction and drop-weight impact tests following Pajunen et al. [2019]. Primary metrics include *peak transmitted force*, *specific energy absorption* (SEA), and *compaction efficiency*.

With initial data in hand, the student will **implement a BO loop** (BoTorch/Ax) that operates directly on experimental measurements. After each iteration the surrogate recommends the next batch of designs; the student coordinates with Student 1 to print and test them, closing the loop. If time permits, lightweight simulations may be added as a low-fidelity data source to accelerate convergence [Mo et al., 2023], but the campaign does not depend on simulation.

Prof. Baird's master's student provides day-to-day co-mentoring in Python, data analysis, and BO implementation. In Year 2, both students mentor newly recruited undergraduates, reinforcing their own understanding while sustaining the research pipeline.

Mentoring outcomes: The student will develop skills in experimental mechanics (test design, data acquisition), scientific computing (Python), and data-driven optimization. They will present results at **ASME IDETC** or **UCUR** and contribute as co-author on a journal manuscript.

5 Mentoring Environment

Each student follows a structured path from guided exploration to independent research ownership, supported by layered mentoring. Because the project centers on hands-on experimentation—designing, printing, and testing physical structures—students engage with tangible problems from day one.

Recruitment

Students will be recruited from courses in mechanics of materials, machine design, kinematics, and compliant mechanisms, targeting sophomores through seniors. The experiment-driven nature of the project lowers the barrier to entry: students need not have prior simulation or programming experience.

Mentoring Structure

Students meet individually with Profs. Hill and Baird each week (30 min) for technical guidance and professional development, and participate in weekly lab group meetings (1 hr) that include progress updates, literature presentations, and skill-building workshops on scientific writing, presentation skills, and research ethics. Prof. Baird's master's student serves as a graduate co-mentor, providing day-to-day guidance in 3D printing, testing, coding, and data interpretation. Students begin with structured tasks—replicating known tensegrity geometries and running standard test protocols—and progress to independent design decisions, culminating in ownership of their research project. Year 1 focuses on skill acquisition and guided execution; Year 2 emphasizes independent hypothesis generation, experimental design, and peer mentoring of newly recruited sophomores.

Student Development Outcomes

Mentoring is structured around concrete milestones: a student's first successful multi-material print, first compression test, first data set fed into the BO loop. *Technical* skills include experimental mechanics, additive manufacturing (CAD, multi-material 3D printing), and data-driven optimization (Python, Bayesian optimization). *Communication* skills develop through lab notebooks, group presentations, conference talks, and co-authoring a peer-reviewed manuscript. *Professional* development includes project management, data-driven decision-making, interdisciplinary collaboration, and preparation for graduate school or industry careers.

6 Expected Research Outcomes

The primary outcome is the **training of 2 undergraduate researchers** with hands-on experience in multi-material 3D printing, mechanical testing, and data-driven optimization. Each undergraduate is expected to have participated in the design and testing of approximately 100+ tensegrity specimens, contributed to **conference presentations** at UCUR and ASME IDETC, and co-authored a peer-reviewed manuscript submitted to the *ASME Journal of Mechanical Design* or *Smart Materials and Structures*. The experimental data set—spanning a wide range of tensegrity-inspired geometries with measured energy-absorption performance—will be published on GitHub for reproducibility. In Year 2, students will mentor incoming undergraduates, developing leadership skills and creating a self-sustaining research group.

7 Potential Impact of Work

This project advances architected material design for energy absorption with applications in **protective equipment** (helmets, body armor), **packaging** (shock-absorbing inserts), and **aerospace** (lightweight impact-resistant structures). The systematically collected experimental data set provides the community with a resource for benchmarking and further optimization. The experiment-driven BO framework demonstrates that undergraduates can drive a closed-loop experimental campaign, establishing a replicable mentoring model for data-driven research. Open-source dissemination of data, code, and tested designs extends impact beyond BYU.

8 Timeline

The project spans two academic years plus two summers (Table 1). **Year 1** focuses on experimental foundations: recruitment and onboarding in Fall, literature review and CAD parameterization through Winter, and a Summer term of full-time research with intensive mentoring where students ramp up fabrication, run systematic compression and impact tests, and begin feeding results into the BO loop. **Year 2** shifts to optimization and dissemination: BO-guided campaigns continue through Fall and Winter with progressively refined designs. Students present at UCUR and ASME IDETC, and a journal manuscript is prepared during Winter and Summer. Experienced students mentor newly recruited undergraduates throughout Year 2.

Task	Y1 F	Y1 W	Y1 Su	Y2 F	Y2 W	Y2 Su
Student recruitment & onboarding	•					
Literature review	•	•				
Tensegrity parameterization & CAD	•	•				
Print trials & process validation	•	•	•			
Multi-material 3D printing campaigns		•	•	•	•	
Compression & impact testing		•	•	•	•	
Bayesian optimization loop			•	•	•	
Conference presentations (UCUR/IDETC)				•	•	
Journal manuscript preparation					•	•
Peer mentoring of new students				•	•	

Table 1: Project timeline across two years (F = Fall, W = Winter, Su = Summer). Summer terms support full-time research with intensive mentoring. Year 2 includes formal peer mentoring, where experienced students supervise newly recruited undergraduates.

9 Budget

Total requested: **$25,000** over 2 years. Per grant guidelines, no funds are allocated to equipment, tuition, or faculty wages.

Category	Description	Year 1	Year 2
Undergraduate wages	2 students, part-time + summer full-time	$10,000	$10,000
Graduate student wages	Partial RA for grad co-mentor	$1,250	$1,250
Supplies	Filament, fasteners, consumables	$750	$250
Travel	Conference registration & travel	$0	$1,500
Annual Total		**$12,000**	**$13,000**

Undergraduate wages ($20,000): Supports 2 undergraduates working part-time (∼10 hrs/week) during fall and winter semesters and full-time (∼40 hrs/week) during summer terms for design, 3D printing, and testing. **Graduate student wages ($2,500):** Prof. Baird's master's student serves as day-to-day co-mentor. **Supplies ($1,000):** PLA/TPU filament (sufficient for 50–100+ specimens), fasteners, poster printing, and other consumables. **Travel ($1,500):** Student presentations at ASME IDETC or UCUR (Year 2).

References

Ada Amendola, Gerardo Carpentieri, Mauricio C. de Oliveira, Robert E. Skelton, and Fernando Fraternali. Experimental investigation of the softening–stiffening response of tensegrity prisms under compressive loading. *Composite Structures*, 117:234–243, 2014. doi: 10.1016/j.compstruct.2014.06.022.

Sebastian Ament, Andrew Witte, Nishant Garg, and Julius Kusuma. Sustainable concrete via Bayesian optimization. *arXiv preprint arXiv:2310.18288*, 2023. URL `https://arxiv.org/abs/2310.18288`. NeurIPS 2023 Workshop on Adaptive Experimental Design and Active Learning in the Real World.

Peter I. Frazier. A tutorial on Bayesian optimization. *arXiv preprint arXiv:1807.02811*, 2018. URL `https://arxiv.org/abs/1807.02811`.

Nava Raj Khatri and Paul F. Egan. Energy absorption of 3D printed ABS and TPU multimaterial honeycomb structures. *3D Printing and Additive Manufacturing*, 11:e840–e850, 2024. doi: 10.1089/3dp.2022.0196.

Changyu Mo, Paris Perdikaris, and Jordan R. Raney. Accelerated design of architected materials with multifidelity Bayesian optimization. *Journal of Engineering Mechanics*, 149(6):04023028, 2023. doi: 10.1061/JENMDT.EMENG-7033.

Kirsti Pajunen, Paul Johanns, Raj Kumar Pal, Julian J. Rimoli, and Chiara Daraio. Design and impact response of 3D-printable tensegrity-inspired structures. *Materials & Design*, 182:107966, 2019. doi: 10.1016/j.matdes.2019.107966.

Bobak Shahriari, Kevin Swersky, Ziyu Wang, Ryan P. Adams, and Nando de Freitas. Taking the human out of the loop: A review of Bayesian optimization. *Proceedings of the IEEE*, 104(1):148–175, 2016. doi: 10.1109/JPROC.2015.2494218.

David Silver, Aja Huang, Chris J. Maddison, Arthur Guez, Laurent Sifre, George van den Driessche, Julian Schrittwieser, Ioannis Antonoglou, Veda Panneershelvam, Marc Lanctot, et al. Mastering the game of Go with deep neural networks and tree search. *Nature*, 529(7587):484–489, 2016. doi: 10.1038/nature16961.

Robert E. Skelton and Mauricio C. de Oliveira. *Tensegrity Systems*. Springer, New York, 2009. doi: 10.1007/978-0-387-74242-7.

Felix Strieth-Kalthoff, Han Hao, Vandana Rathore, Joshua Derasp, Théophile Gaudin, Nicholas H. Angello, Martin Seifrid, Ekaterina Trushina, Mason Guy, Junliang Liu, et al. Delocalized, asynchronous, closed-loop discovery of organic laser emitters. *Science*, 384(6697):eadk9227, 2024. doi: 10.1126/science.adk9227.

Cornel Sultan. Tensegrity: 60 years of art, science, and engineering. *Advances in Applied Mechanics*, 43:69–145, 2009. doi: 10.1016/S0065-2156(09)43002-3.

Haitao Ye, Qingjiang Liu, Jianxiang Cheng, Honggeng Li, Bingcong Jian, Rong Wang, Zechu Sun, Yang Lu, and Qi Ge. Multimaterial 3D printed self-locking thick-panel origami metamaterials. *Nature Communications*, 14:1607, 2023. doi: 10.1038/s41467-023-37343-w.

Qicheng Zhang, Dayi Zhang, Yousef Dobah, Fabrizio Scarpa, Fernando Fraternali, and Robert E. Skelton. Tensegrity cell mechanical metamaterial with metal rubber. *Applied Physics Letters*, 113(3):031906, 2018. doi: 10.1063/1.5040850.

Biographical Sketches

Jeffrey R. Hill — Principal Investigator

A. Personal Statement

I am an Associate Professor of Mechanical Engineering at Brigham Young University with expertise in tensegrity structures, smart materials, and experimental mechanics. My research group has recently published on tensegrity robot locomotion, cable tension interactions, and 3D-printed tension networks, providing direct expertise relevant to this proposal. I have a strong record of mentoring undergraduate and graduate students through hands-on research, which aligns with the goals of the Mentored Research Grant program.

1. Layer B, Denning H, Hill J. Control and locomotion of tensegrity robots through manipulation of the center of mass. *Robotica*. 2024. `https://doi.org/10.1017/S0263574724001139`
2. Masmeijer T, Swain C, Hill J, Habtour E. Programming tension in 3D printed networks inspired by spiderwebs. *Materials & Design*. 2025. `https://doi.org/10.1016/j.matdes.2025.115281`

B. Positions, Scientific Appointments, and Honors

Education/Training

Institution	Degree	Completion Date	Field of Study
Pennsylvania State University	Ph.D.	2011	Mechanical Engineering
Brigham Young University	M.S.	2005	Mechanical Engineering
Brigham Young University	B.S.	2004	Mechanical Engineering

Positions and Employment

- 2021–present: Associate Professor, Dept. of Mechanical Engineering, Brigham Young University
- 2016–2021: Principal Member of the Technical Staff, Sandia National Laboratories
- 2011–2016: Senior Member of the Technical Staff, Sandia National Laboratories
- 2008–2011: Visiting Research Investigator, University of Michigan

C. Contributions to Science

1. Tensegrity structures and robots. My group investigates design, construction, and control of tensegrity systems. We have demonstrated simplified construction methods, characterized cable tension interactions, and developed locomotion strategies for tensegrity robots.

1. Layer B, Denning H, Hill J. Control and locomotion of tensegrity robots through manipulation of the center of mass. *Robotica*. 2024. `https://doi.org/10.1017/S0263574724001139`
2. Denning H, Bullinger A, Hill J. Tuning Tensegrity: Simplified Construction and Cable Tension Interactions. *ASME SMASIS 2024*. `https://doi.org/10.1115/SMASIS2024-140381`
3. Brown A, Swain C, Usevitch N, Hill J. Development of a Continuous Cable Tensegrity. *ASME SMASIS 2024*. `https://doi.org/10.1115/SMASIS2024-140185`

2. Advanced manufacturing and structural design. I have contributed to 3D-printed tension networks inspired by spiderwebs and encapsulation techniques for protecting electronic components, demonstrating expertise in additive manufacturing and structural characterization.

1. Masmeijer T, Swain C, Hill J, Habtour E. Programming tension in 3D printed networks inspired by spiderwebs. *Materials & Design*. 2025. `https://doi.org/10.1016/j.matdes.2025.115281`
2. Hill J, Chen A, Wilson N, Boll C, O'Bannon M, Trentman D. Microbead Encapsulation for Protection of Electronic Components. *J. Electronic Packaging*. 2025;147(021007). `https://doi.org/10.1115/1.4067650`

3. Smart materials and structural optimization. My earlier work focused on actuator grouping optimization and piezoelectric fiber composite analysis for adaptive structures, establishing a foundation in computational and experimental approaches to structural systems.

1. Hill J, Wang K. Development of the En Masse Elimination Algorithm for Actuator Grouping Optimization. *J. Intelligent Material Systems and Structures*. 2011;22(11):1203–1212. `https://doi.org/10.1177/1045389X11417651`
2. Kim J, Hill J, Wang K. An asymptotic approach for the analysis of piezoelectric fiber composite beams. *Smart Materials and Structures*. 2011;20(2):025023.
3. Hill J, Wang K, Fang H, Quijano U. Actuator grouping optimization on flexible space reflectors. *SPIE Active and Passive Smart Structures*. 2011. `https://doi.org/10.1117/12.880086`

Sterling G. Baird — Co-Principal Investigator

A. Personal Statement

I am an Assistant Professor of Mechanical Engineering at Brigham Young University specializing in Bayesian optimization, self-driving laboratories, and data-driven materials discovery. My work on the Honegumi interface and high-dimensional Bayesian optimization directly supports the optimization component of this proposal. I have extensive experience mentoring undergraduate researchers and developing accessible open-source tools that lower barriers to adopting advanced optimization methods.

1. Baird SG, Falkowski AR, Sparks TD. Honegumi: An Interface for Accelerating the Adoption of Bayesian Optimization in the Experimental Sciences. 2025. `https://doi.org/10.48550/arXiv.2502.06815`
2. Tom G, Schmid SP, Baird SG, et al. Self-Driving Laboratories for Chemistry and Materials Science. *Chemical Reviews*. 2024;124(16):9633–9732. `https://doi.org/10.1021/acs.chemrev.4c00055`

B. Positions, Scientific Appointments, and Honors

Education/Training

Institution	Degree	Completion Date	Field of Study
University of Utah	Ph.D.	2023	Materials Science and Engineering
Brigham Young University	M.S.	2021	Mechanical Engineering
Brigham Young University	B.S.	2018	Applied Physics

Positions and Employment

- 2025–present: Assistant Professor, Dept. of Mechanical Engineering, Brigham Young University
- 2025: Principal Investigator and Staff Scientist, Acceleration Consortium, University of Toronto
- 2023–2025: Director of Training and Programs, Acceleration Consortium, University of Toronto

Honors

- Guest Editor, *npj Computational Materials*
- Journal reviewer for *Nature Communications*, *Digital Discovery*, *JOSS*

C. Contributions to Science

1. Bayesian optimization for materials and experimental sciences. I developed Honegumi, an interactive interface for generating Bayesian optimization scripts, and demonstrated high-dimensional BO applied to materials property prediction and formulation optimization.

1. Baird SG, Falkowski AR, Sparks TD. Honegumi: An Interface for Accelerating the Adoption of Bayesian Optimization in the Experimental Sciences. 2025. `https://doi.org/10.48550/arXiv.2502.06815`
2. Baird SG, Liu M, Sparks TD. High-Dimensional Bayesian Optimization of 23 Hyperparameters over 100 Iterations for an Attention-Based Network to Predict Materials Property. *Computational Materials Science*. 2022;211:111505. `https://doi.org/10.1016/j.commatsci.2022.111505`
3. Baird SG, Hall JR, Sparks TD. Compactness Matters: Improving Bayesian Optimization Efficiency of Materials Formulations through Invariant Search Spaces. *Computational Materials Science*. 2023;224:112134. `https://doi.org/10.1016/j.commatsci.2023.112134`
4. Baird SG, et al. Bayesian Optimization Hackathon for Chemistry and Materials. 2025. `https://doi.org/10.26434/chemrxiv-2025-dzh5z`

2. Self-driving laboratories. I co-authored the definitive review of self-driving labs for chemistry and materials science and developed a low-cost "Hello World" demonstration platform, making autonomous experimentation accessible to a broad audience.

1. Tom G, Schmid SP, Baird SG, et al. Self-Driving Laboratories for Chemistry and Materials Science. *Chemical Reviews*. 2024;124(16):9633–9732. `https://doi.org/10.1021/acs.chemrev.4c00055`
2. Baird SG, Sparks TD. Building a "Hello World" for Self-Driving Labs: The Closed-loop Spectroscopy Lab Light-mixing Demo. *STAR Protocols*. 2023;4(2):102329. `https://doi.org/10.1016/j.xpro.2023.102329`
3. Lo S, Baird SG, et al. Review of Low-Cost Self-Driving Laboratories in Chemistry and Materials Science: The "Frugal Twin" Concept. *Digital Discovery*. 2024;3(5):842–868. `https://doi.org/10.1039/D3DD00223C`

3. Community building and open-source tools for data-driven research. I organized a

Bayesian optimization hackathon (120 participants, 62 organizations, 14 countries) and developed open-source tools that lower the barrier to adopting data-driven methods in experimental research. I have mentored 8 undergraduate research assistants at BYU and 12+ staff and students at the Acceleration Consortium.

1. Baird SG, et al. Bayesian Optimization Hackathon for Chemistry and Materials. 2025. `https://doi.org/10.26434/chemrxiv-2025-dzh5z`
2. Seegmiller CC, Baird SG, Sayeed HM, Sparks TD. Discovering Chemically Novel, High-Temperature Superconductors. *Computational Materials Science*. 2023.